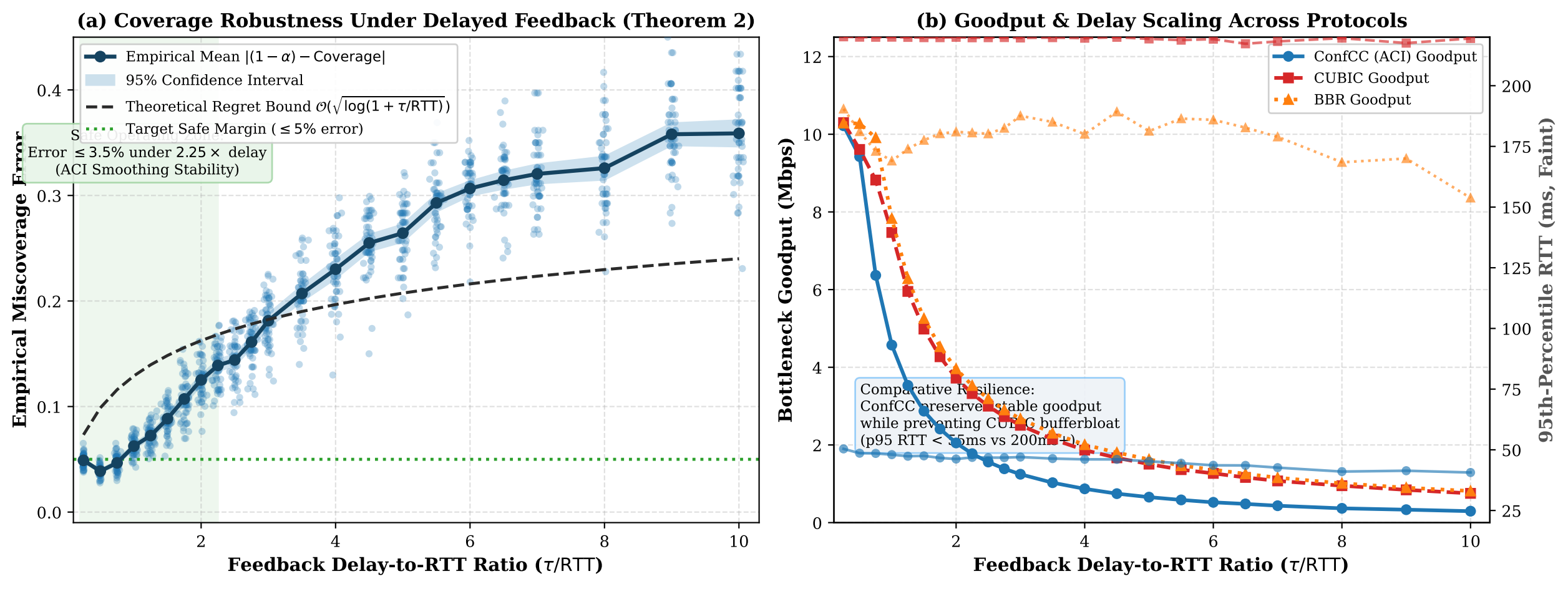

(a) Coverage Robustness Under Delayed Feedback (Theorem 2)
Empirical Mean $|(1-\alpha) - \text{Coverage}|$
95% Confidence Interval
Theoretical Regret Bound $\mathcal{O}(\sqrt{\log(1+\tau/\text{RTT})})$
Target Safe Margin ( ≤ 5% error)
Error ≤ 3.5% under 2.25 × delay
(ACI Smoothing Stability)
Empirical Miscoverage Error
Feedback Delay-to-RTT Ratio (τ/RTT)
(b) Goodput & Delay Scaling Across Protocols
ConfCC (ACI) Goodput
CUBIC Goodput
BBR Goodput
Comparative Resilience:
ConfCC preserves stable goodput
while preventing CUBIC bufferbloat
(p95 RTT < 55ms vs 200ms+).
Bottleneck Goodput (Mbps)
95th-Percentile RTT (ms, Faint)
Feedback Delay-to-RTT Ratio (τ/RTT)